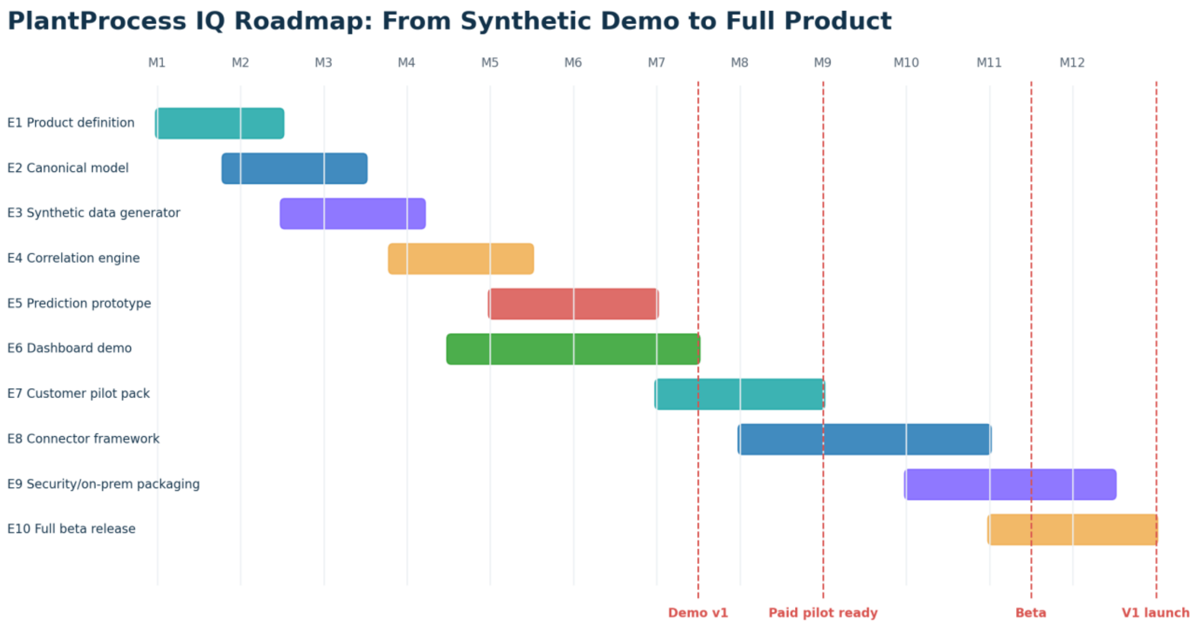
PlantProcess IQ

Full Product Plan: From Scratch to Demo, Pilot and V1



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1. Planning Objective

This plan converts the PlantProcess IQ concept into a practical roadmap. The goal is to build a demo and pilot-ready product without using any confidential employer/customer data. The first 12 months should focus on synthetic data, a generic model, one strong steel-inspired example, and customer validation before any full product commitment.

Planning rule
Do not build the full platform before selling the problem. Build a credible synthetic demo, run discovery calls, sell a data readiness assessment, then sell a pilot.

2. Roadmap Overview

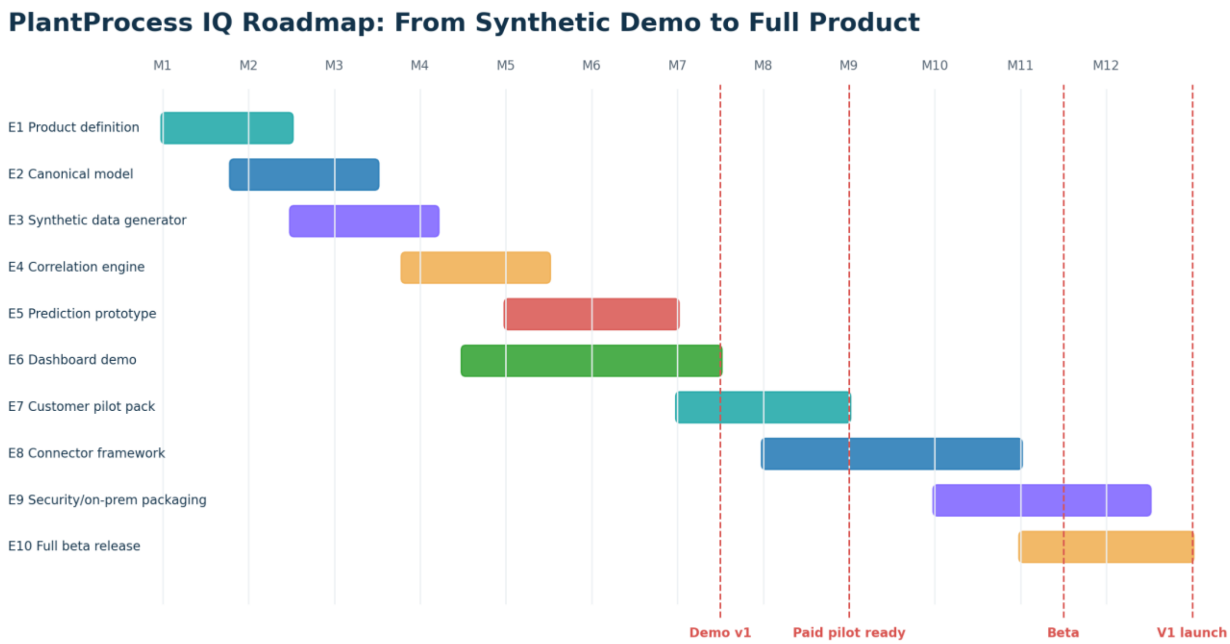

Figure 1 - Roadmap from product definition to V1 launch.

3. Milestones

Milestone	Target Timing	Definition of Done
M0 - Concept locked	Week 1	Product name, target customer, first use case, value proposition and non-goals defined.
M1 - Technical blueprint ready	Week 2	Architecture, data model, stack, MVP scope and demo story finalized.
M2 - Synthetic data generator ready	Week 4-6	100k+ synthetic records generated for process, quality, defects and genealogy.
M3 - Data model and ETL ready	Week 6-8	Canonical schema implemented and synthetic source data mapped into it.
M4 - Correlation demo ready	Week 10-12	Dashboard shows top risk contributors and defect correlations.
M5 - Prediction demo ready	Month 4	Baseline ML model predicts risk on synthetic test set and explains top drivers.
M6 - Customer demo pack ready	Month 5-6	Proposal, demo script, sample report and dashboard screenshots ready.

M7 - First discovery calls	Month 6-7	10-20 customer/recruiter/domain conversations completed.
M8 - Pilot-ready product	Month 8-9	Can accept customer sample data and run data readiness/correlation pilot.
M9 - Beta V1	Month 10-12	Connector framework, security baseline, reporting and deployment package ready.

4. Epic Map

Epic ID	Epic Name	Goal
E1	Product Definition & Positioning	Define product boundary, target industries, target personas and first offer.
E2	Generic Manufacturing Data Model	Build domain-neutral entities and configuration strategy.
E3	Synthetic Data Generator	Generate realistic manufacturing data without confidential information.
E4	Data Ingestion & Mapping	Load source tables and map them to canonical entities.
E5	Data Quality & Genealogy Engine	Detect broken data and build material/batch genealogy.
E6	Correlation & Analytics Engine	Find relationships between process parameters and quality outcomes.
E7	Prediction & Explainability	Train baseline ML models and explain high-risk cases.
E8	Dashboard & Frontend	Create Qlik/Power BI dashboard and later React product UI.
E9	Reporting & AI Assistant	Generate investigation reports and natural-language explanations.
E10	Security, Packaging & Deployment	Prepare on-prem/private deployment and customer data controls.
E11	Sales Assets & Customer Validation	Create proposal, demo script, one-page offer and discovery process.

5. Jira-Style Backlog

Ticket	Area	Task	Priority	Estimate
E1-01	Product	Define product name and positioning statement	High	1 day
E1-02	Product	Define first industry-neutral use case and steel-specific demo story	High	2 days
E1-03	Product	Define non-goals: no MES replacement, no full Level 2, no confidential data	High	1 day
E2-01	Data Model	Create ERD for plant, line, operation, material unit, equipment, parameter, quality event	High	3 days
E2-02	Data Model	Design material genealogy table and parent-child relationships	High	3 days
E2-03	Data Model	Design configurable parameter catalog and defect catalog	High	2 days
E3-01	Synthetic Data	Generate steel route: EAF/LF -> Caster -> HSM -> Quality	High	5 days

E3-02	Synthetic Data	Add hidden defect patterns for ML validation	High	3 days
E3-03	Synthetic Data	Add generic manufacturing examples: batch, equipment, quality result	Medium	3 days
E4-01	Ingestion	Create source import format for CSV/Excel/SQL export	High	4 days
E4-02	Ingestion	Implement mapping configuration from source columns to canonical model	High	5 days
E5-01	Data Quality	Implement missing ID, duplicate event and timestamp gap checks	High	4 days
E5-02	Genealogy	Build material genealogy search API	High	5 days
E6-01	Analytics	Build correlation matrix by defect and parameter	High	4 days
E6-02	Analytics	Build defect-rate-by-parameter-bin analysis	High	3 days
E7-01	ML	Train baseline random forest/gradient boosting model	High	5 days
E7-02	ML	Add feature importance and explanation text	High	4 days
E8-01	Dashboard	Create Qlik/Power BI dashboard mockup	High	5 days
E8-02	Frontend	Create React UI skeleton for material risk and genealogy	Medium	7 days
E9-01	Reports	Generate PDF investigation report	Medium	4 days
E10-01	Security	Define roles, permissions, audit logs and on-prem deployment assumptions	High	4 days
E11-01	Sales	Create one-page offer and sales deck/doc	High	3 days
E11-02	Validation	Prepare 20 discovery questions and outreach target list	High	2 days

6. Sprint Plan

Sprint	Duration	Goal	Demo Output
Sprint 1	2 weeks	Product scope, architecture, data model	Technical blueprint and ERD draft
Sprint 2	2 weeks	Synthetic data generator	Generated process/quality/genealogy dataset
Sprint 3	2 weeks	Database schema and ingestion	Data loaded into canonical model
Sprint 4	2 weeks	Data quality and genealogy	Search material and show process history
Sprint 5	2 weeks	Correlation engine	Correlation dashboard and top contributors
Sprint 6	2 weeks	Prediction model	Risk scoring and feature importance
Sprint 7	2 weeks	Dashboard/report pack	Demo dashboard and sample

			customer report
Sprint 8	2 weeks	Sales demo readiness	Recorded demo, proposal, discovery script

7. When Can You Show a Demo?

Earliest credible demo

After 8-12 weeks, you can show a synthetic demo: generated plant data, genealogy, defect correlation, risk dashboard and sample explanation. This is enough for discovery calls, not yet for a paid production pilot.

Demo Level	Timing	What It Includes	Can Show To
Concept demo	Week 2	Architecture, mockups, value proposition	Trusted contacts / advisors
Synthetic data demo	Week 6	Generated data and basic genealogy	Technical peers / recruiters
Analytics demo	Week 10-12	Correlation dashboard and defect patterns	Potential customers
Prediction demo	Month 4	Risk model and explanation	Potential pilot customers
Pilot-ready demo	Month 6-8	Data readiness workflow and import template	Paying pilot prospects
Beta product	Month 10-12	Connector framework, reports, deployment package	First paying customer

8. Customer Validation Plan

1. Prepare a one-page offer: Process-to-Quality Correlation Diagnostic.
2. Identify target personas: plant quality manager, MES manager, process engineer, manufacturing IT lead, digital manufacturing manager, system integrator.
3. Send 50 focused outreach messages with one simple question: “Do you have difficulty linking process data to quality defects?”
4. Run discovery calls without selling the full product.
5. Ask for examples of their defect investigation process, data sources and pain points.
6. Refine the product around repeated pain, not around personal assumptions.
7. Only then offer a paid data readiness assessment or diagnostic pilot.

9. Risk Register

Risk	Impact	Mitigation
Scope becomes too broad	Product never finishes	Start with one route, one defect family, one dashboard.
No real customer data	Model cannot prove value	Use synthetic demo first, then sell data readiness assessment.
False correlation interpreted as root cause	Loss of trust	Use careful wording: suspected contributor, not guaranteed cause.
Security restrictions block access	Pilot delayed	Offer CSV/SQL export first, on-prem later.
Too much Qlik dependency	Product becomes only dashboard	Keep intelligence engine separate from BI layer.
Too much ML complexity	Hard to explain	Start with correlation and interpretable models.
Conflict with employer/client data	Legal risk	Use only synthetic data and customer-approved data; keep all work independent.
Customer expects full MES	Wrong expectation	Position as intelligence layer, not MES replacement.

10. Definition of Full Version

- Multi-source data ingestion: CSV/Excel, SQL Server, Oracle, PostgreSQL and API-based import.
- Configurable canonical model for different industries and material/batch types.
- Genealogy viewer for material/batch/product traceability.
- Data quality engine with reusable rules and issue reporting.
- Correlation engine for parameter-to-defect analysis.
- Prediction engine with model registry, validation, risk scoring and explainability.
- Dashboard layer with Qlik/Power BI export and React product UI.
- Investigation report generator with evidence, charts, suspected contributors and next actions.
- On-prem/private cloud deployment package with RBAC, audit logs and secure configuration.
- Sales/pilot kit: proposal, demo script, data readiness checklist and sample report.

11. Suggested First Repository Structure

Folder / Project	Purpose
src/PlantProcess.Api	.NET backend API for dashboard and product workflows
src/PlantProcess.Domain	Canonical domain model and business rules
src/PlantProcess.Infrastructure	Database, connectors, import jobs and external integrations
src/PlantProcess.Workers	Background ingestion, analytics and report jobs
ml/	Python notebooks/scripts for correlation, feature engineering and model training
ui/	React frontend product UI
dashboards/	Qlik/Power BI dashboard prototypes and exported screenshots
data/synthetic/	Generated synthetic datasets only
docs/	Architecture, data dictionary, product proposal, pilot reports
deploy/	Docker compose, local deployment and customer pilot setup

12. References

- Microsoft .NET support policy and release lifecycle: https://dotnet.microsoft.com/en-us/platform/support/policy
- ML.NET documentation: https://learn.microsoft.com/en-us/dotnet/machine-learning/
- scikit-learn documentation: https://scikit-learn.org/
- Qlik Sense analytic connections / Server-Side Extension: https://help.qlik.com/en-US/sense-developer/November2025/Content/Sense_Helpsites/AnalyticConnections.htm
- PostgreSQL declarative partitioning: https://www.postgresql.org/docs/current/ddl-partitioning.html
- PostgreSQL JSON/JSONB types: https://www.postgresql.org/docs/current/datatype-json.html
- ONNX Runtime documentation: https://onnxruntime.ai/docs/